DISCRIMINANTS AND RAMIFIED PRIMES

KEITH CONRAD

1. Introduction

A prime number p is said to be *ramified* in a number field K if the prime ideal factorization

$$(p) = p\mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_g^{e_g} \tag{1.1}$$

has some e_i greater than 1. If every e_i equals 1, we say p is *unramified* in K.

Example 1.1. In $\mathbf{Z}[i]$, the only prime that ramifies is 2: $(2) = (1+i)^2$.

Example 1.2. Let $K = \mathbf{Q}(\alpha)$ where α is a root of $f(T) = T^3 - 9T - 6$. Then $6 = \alpha^3 - 9\alpha = \alpha(\alpha-3)(\alpha+3)$. For $m \in \mathbf{Z}$, $\alpha + m$ has minimal polynomial $f(T-m)$ in $\mathbf{Q}[T]$, so $\mathrm{N}_{K/\mathbf{Q}}(\alpha+m) = -f(-m) = m^3 - 9m + 6$ and the principal ideal $(\alpha - m)$ has norm

$$\mathrm{N}(\alpha - m) = |m^3 - 9m + 6|.$$

Therefore $\mathrm{N}(\alpha) = 6$, $\mathrm{N}(\alpha - 3) = 6$, and $\mathrm{N}(\alpha + 3) = 6$. It follows that $(\alpha) = \mathfrak{p}_2\mathfrak{p}_3$, $(\alpha - 3) = \mathfrak{p}_2'\mathfrak{p}_3$, and $(\alpha+3) = \mathfrak{p}_2'\mathfrak{p}_3$ (so, in particular, $\alpha + 3$ and $\alpha - 3$ are unit multiples of each other). Thus

$$(2)(3) = (6) = (\alpha)(\alpha-3)(\alpha+3) = \mathfrak{p}_2\mathfrak{p}_2'^2\mathfrak{p}_3^3,$$

so $(2) = \mathfrak{p}_2^2\mathfrak{p}_2'$ and $(3) = \mathfrak{p}_3^3$. This shows 2 and 3 are ramified in K. Note that one of the exponents in the factorization of (2) exceeds 1 while the other equals 1.

One way to think about ramified primes is in terms of the ring structure of $\mathcal{O}_K/(p)$. By (1.1) and the Chinese Remainder Theorem,

$$\mathcal{O}_K/(p) \cong \mathcal{O}_K/\mathfrak{p}_1^{e_1} \times \cdots \times \mathcal{O}_K/\mathfrak{p}_g^{e_g}. \tag{1.2}$$

If some e_i is greater than 1, then the quotient ring $\mathcal{O}_K/\mathfrak{p}_i^{e_i}$ has a nonzero nilpotent element (use the reduction modulo $\mathfrak{p}_i^{e_i}$ of any element of $\mathfrak{p}_i - \mathfrak{p}_i^{e_i}$), so the product ring (1.2) has a nonzero nilpotent element. If each e_i equals 1, then $\mathcal{O}_K/(p)$ is a product of (finite) fields, and a product of fields has no nonzero nilpotent elements. Thus, p ramifies in K if and only if $\mathcal{O}_K/(p)$ has a nonzero nilpotent element.

In Sections 2 and 3 we will prove the following result of Dedekind [2], which characterizes the prime numbers ramifying in a number field K in terms of its discriminant.[1]

Theorem 1.3 (Dedekind)**.** *For a number field K, a prime p ramifies in K if and only if p divides the integer* $\mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K)$.

Recall $\mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K) := \mathrm{disc}_{K/\mathbf{Q}}(\alpha_1, \ldots, \alpha_n)$ where $\{\alpha_1, \ldots, \alpha_n\}$ is any $\mathbf{Z}$-basis of $\mathcal{O}_K$. The generalization of Theorem 1.3 to a relative extension of number fields is in Section 4.

[1] In [1, pp. 36-37], Dedekind said he announced this result for the first time on September 20, 1871.

2. A special case

We will first consider Theorem 1.3 in the special case that there is an $\alpha \in \mathcal{O}_K$ such that $K = \mathbf{Q}(\alpha)$ and $p \nmid [\mathcal{O}_K : \mathbf{Z}[\alpha]]$. (If $\mathcal{O}_K = \mathbf{Z}[\alpha]$ we can use the same α for all p.) The treatment of the general case in Section 3 will not rely on the case that $p \nmid [\mathcal{O}_K : \mathbf{Z}[\alpha]]$ for some α, but this special case is technically simpler. We will just sketch the basic ideas behind it.

Proof. Assume for a prime p that there's $\alpha \in \mathcal{O}_K$ such that $K = \mathbf{Q}(\alpha)$ and $p \nmid [\mathcal{O}_K : \mathbf{Z}[\alpha]]$. Let $f(T)$ be the minimal polynomial of α over $\mathbf{Q}$, so $f(T)$ is monic in $\mathbf{Z}[T]$. and the Dedekind–Kummer theorem tells us that the way $p\mathcal{O}_K$ factors into prime ideals matches the way $f(T)$ mod p factors into monic irreducibles in $(\mathbf{Z}/p\mathbf{Z})[T]$.

Writing the prime ideal factorization of $p\mathcal{O}_K$ as in (1.1), by definition p ramifies in $\mathcal{O}_K$ if and only if some $e_i > 1$. The factorization of the mod p reduction $\overline{f}(T)$ in $(\mathbf{Z}/p\mathbf{Z})[T]$ matches (1.1), in the sense that

$$\overline{f}(T) = \pi_1^{e_1} \cdots \pi_g^{e_g}$$

for some distinct monic irreducibles $\pi_i \in (\mathbf{Z}/p\mathbf{Z})[T]$.

The irreducible polynomials $\pi_i(T)$'s are separable (all irreducibles over a finite field are separable), so some $e_i > 1$ if and only if $\overline{f}(T)$ has a repeated root in a splitting field over $\mathbf{Z}/p\mathbf{Z}$. This is equivalent to $\overline{f}(T)$ having discriminant 0, so p ramifies in $\mathcal{O}_K$ if and only if $\mathrm{disc}(\overline{f}) = 0$ in $\mathbf{Z}/p\mathbf{Z}$.

Since the discriminant of a monic polynomial is a universal polynomial in its coefficients (consider the quadratic case, where $T^2 + bT + c$ has discriminant $b^2 - 4c$), discriminants of monic polynomials behave well under reduction: $\mathrm{disc}(f(T) \bmod p) = \mathrm{disc}(f(T)) \bmod p$. Therefore $\mathrm{disc}(\overline{f}(T)) = 0$ in $\mathbf{Z}/p\mathbf{Z}$ if and only if $\mathrm{disc}(f) \equiv 0 \bmod p$. Thus p ramifies in $\mathcal{O}_K$ if and only if $p \mid \mathrm{disc}(f)$. Since

$$\mathrm{disc}(f) = \mathrm{disc}(\mathbf{Z}[\alpha]) = [\mathcal{O}_K : \mathbf{Z}[\alpha]]^2 \, \mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K)$$

and $p \nmid [\mathcal{O}_K : \mathbf{Z}[\alpha]]$, we have $p \mid \mathrm{disc}(f)$ if and only if $p \mid \mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K)$. □

3. The general case

To prove Theorem 1.3 for every prime number p, even if $p \mid [\mathcal{O}_K : \mathbf{Z}[\alpha]]$ for all $\alpha \in \mathcal{O}_K$ such that $K = \mathbf{Q}(\alpha)$,[2] we will examine discriminants of ring extensions to show computing the discriminant commutes with reduction mod p: $\mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K) \bmod p = \mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/(p))$. Then we will use (1.2) to write $\mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/(p))$ as a product of discriminants of rings of type $\mathcal{O}_K/\mathfrak{p}^e$ and compute the discriminants of these particular rings.

Definition 3.1. Let A be a commutative ring and B be a ring extension of A that is a finite free A-module:

$$B = Ae_1 \oplus \cdots \oplus Ae_n.$$

Then we set

$$\mathrm{disc}_A(e_1, \ldots, e_n) = \det(\mathrm{Tr}_{B/A}(e_i e_j)) \in A.$$

[2]Dedekind wrote in [1, p. 37] that he first proved Theorem 1.3 in the special case of Section 2 above and was able to prove the general case only after many unsuccessful attempts. He presented the special case first in [2, §3] since it's much simpler than the method he used for the general case in [2, §§4-6].

Remark 3.2. The discriminant of a basis is an algebraic concept of "volume". To explain this viewpoint, we should think about $\mathrm{Tr}_{B/A}(xy)$ as an analogue of the dot product $v \cdot w$ in $\mathbf{R}^n$. For a basis $v_1, \ldots, v_n$ in $\mathbf{R}^n$, the ordinary Euclidean volume of the parallelotope

$$\left\{ \sum_{i=1}^{n} a_i v_i : 0 \leq a_i \leq 1 \right\}$$

having edges v_i is $\sqrt{|\det(v_i \cdot v_j)|}$. The discriminant of an A-basis of B uses the A-valued pairing $\langle x, y \rangle = \mathrm{Tr}_{B/A}(xy)$ on B in place of the $\mathbf{R}$-valued dot product on $\mathbf{R}^n$ and we just drop the absolute value and the square root when we make the algebraic analogue.

How are the discriminants of two A-module bases for B related? Pick a second basis $e_1', \ldots, e_n'$ of B as an A-module. Then

$$e_i' = \sum_{j=1}^{n} a_{ij} e_j,$$

where $a_{ij} \in A$ and the change of basis matrix (a_{ij}) has determinant in $A^\times$. Then

$$\begin{aligned} \mathrm{Tr}_{B/A}(e_i' e_j') &= \mathrm{Tr}_{B/A}\left(\sum_{k=1}^{n} a_{ik} e_k \sum_{\ell=1}^{n} a_{j\ell} e_\ell \right) \\ &= \sum_{k=1}^{n} \sum_{\ell=1}^{n} a_{ik} \, \mathrm{Tr}_{B/A}(e_k e_\ell) a_{j\ell}, \end{aligned}$$

so

$$(\mathrm{Tr}_{B/A}(e_i' e_j')) = (a_{ij})(\mathrm{Tr}_{B/A}(e_i e_j))(a_{ij})^\top.$$

Therefore

$$\mathrm{disc}_A(e_1', \ldots, e_n') = (\det(a_{ij}))^2 \, \mathrm{disc}_A(e_1, \ldots, e_n).$$

We set

$$\mathrm{disc}_A(B) = \mathrm{disc}_A(e_1, \ldots, e_n) \in A$$

for any A-module basis $\{e_1, \ldots, e_n\}$ of B. It is well-defined up to a unit square. In particular, the condition $\mathrm{disc}_A(B) = 0$ is independent of the choice of basis.

Given a number field K, ramification of the prime p in K has been linked to the structure of the ring $\mathcal{O}_K/(p)$ in Section 1. Let's look at the discriminant of this ring over $\mathbf{Z}/p\mathbf{Z}$. Letting K have degree n over $\mathbf{Q}$, the ring $\mathcal{O}_K$ is a free rank-n $\mathbf{Z}$-module, say

$$\mathcal{O}_K = \bigoplus_{i=1}^{n} \mathbf{Z}\omega_i.$$

Reducing both sides modulo p,

$$\mathcal{O}_K/(p) = \bigoplus_{i=1}^{n} (\mathbf{Z}/p\mathbf{Z})\overline{\omega}_i,$$

so $\mathcal{O}_K/(p)$ is a vector space over $\mathbf{Z}/p\mathbf{Z}$ of dimension n. The discriminant of $\mathcal{O}_K$ is $\mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K)$. The next lemma says reduction modulo p commutes (in a suitable sense) with the formation of discriminants.

Lemma 3.3. *Choosing bases appropriately for $\mathcal{O}_K$ and $\mathcal{O}_K/(p)$,*

$$\mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K) \bmod p = \mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/(p)).$$

Proof. Pick a $\mathbf{Z}$-basis $\omega_1, \ldots, \omega_n$ of $\mathcal{O}_K$. The reductions $\overline{\omega}_1, \ldots, \overline{\omega}_n$ in $\mathcal{O}_K/(p)$ are a $\mathbf{Z}/p\mathbf{Z}$-basis, so the multiplication matrix $[m_x]$ for any $x \in \mathcal{O}_K$, with respect to the basis $\{\omega_i\}$, reduces modulo p to the multiplication matrix $[m_{\overline{x}}]$ for $\overline{x}$ on $\mathcal{O}_K/(p)$ with respect to the basis $\{\overline{\omega}_i\}$. Therefore

$$\mathrm{Tr}_{(\mathcal{O}_K/(p))/(\mathbf{Z}/p\mathbf{Z})}(\overline{x}) = \mathrm{Tr}(m_{\overline{x}}) = \mathrm{Tr}(m_x) \bmod p = \mathrm{Tr}_{\mathcal{O}_K/\mathbf{Z}}(x) \bmod p.$$

Thus, the mod p reduction of the matrix $(\mathrm{Tr}_{\mathcal{O}_K/\mathbf{Z}}(\omega_i\omega_j))$ is $(\mathrm{Tr}_{(\mathcal{O}_K/(p))/(\mathbf{Z}/p\mathbf{Z})}(\overline{\omega}_i\overline{\omega}_j))$. Now take determinants. □

Lemma 3.4. *Let A be a commutative ring and B_1 and B_2 be commutative ring extensions of A that are each finite free A-modules. Then, choosing A-module bases appropriately,*

$$\mathrm{disc}_A(B_1 \times B_2) = \mathrm{disc}_A(B_1)\,\mathrm{disc}_A(B_2).$$

Proof. Pick A-module bases for B_1 and B_2:

$$B_1 = \bigoplus_{i=1}^{m} Ae_i, \quad B_2 = \bigoplus_{j=1}^{n} Af_j.$$

As an A-module basis for $B_1 \times B_2$ we will use the $m+n$ elements $e_1, \ldots, e_m, f_1, \ldots, f_n$. Since $e_if_j = 0$ in $B_1 \times B_2$, the matrix whose determinant is $\mathrm{disc}_A(B_1 \times B_2)$ is a block diagonal matrix

$$\begin{pmatrix} (\mathrm{Tr}_{(B_1\times B_2)/A}(e_ie_k)) & O \\ O & (\mathrm{Tr}_{(B_1\times B_2)/A}(f_jf_\ell)) \end{pmatrix}.$$

For any $x \in B_1$, multiplication by x on $B_1 \times B_2$ kills the B_2 component and acts on the B_1-component in the way x multiplies on B_1, so a matrix for multiplication by x on $B_1 \times B_2$ is a matrix whose upper left block is a matrix for multiplication by x on B_1 and other blocks are 0. Thus

$$\mathrm{Tr}_{(B_1\times B_2)/A}(x) = \mathrm{Tr}_{B_1/A}(x) \quad \text{for } x \in B_1.$$

Similarly, $\mathrm{Tr}_{(B_1\times B_2)/A}(x) = \mathrm{Tr}_{B_2/A}(x)$ for $x \in B_2$. Thus

$$\begin{pmatrix} (\mathrm{Tr}_{(B_1\times B_2)/A}(e_ie_k)) & O \\ O & (\mathrm{Tr}_{(B_1\times B_2)/A}(f_jf_\ell)) \end{pmatrix} = \begin{pmatrix} (\mathrm{Tr}_{B_1/A}(e_ie_k)) & O \\ O & (\mathrm{Tr}_{B_2/A}(f_jf_\ell)) \end{pmatrix},$$

and taking determinants gives

$$\mathrm{disc}_A(B_1 \times B_2) = \mathrm{disc}_A(B_1)\,\mathrm{disc}_A(B_2).$$ □

Now we prove Theorem 1.3.

Proof. We have $p \mid \mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K)$ if and only if $\mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K) \equiv 0 \bmod p$. By Lemma 3.3

$$\mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K) \bmod p = \mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/(p)),$$

so $p \mid \mathrm{disc}_{\mathbf{Z}}(\mathcal{O}_K)$ if and only if $\mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/(p)) = \overline{0}$ in $\mathbf{Z}/p\mathbf{Z}$.

In (1.2), each factor $\mathcal{O}_K/\mathfrak{p}_i^{e_i}$ is a $\mathbf{Z}/p\mathbf{Z}$-vector space since $p \in \mathfrak{p}_i^{e_i}$. Using (1.2) and Lemma 3.4,

$$\mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/(p)) = \prod_{i=1}^{g} \mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/\mathfrak{p}_i^{e_i}).$$

Therefore we need to show for any prime number p and prime-power ideal $\mathfrak{p}^e$ such that $\mathfrak{p}^e \mid (p)$ that $\mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/\mathfrak{p}^e)$ is $\overline{0}$ in $\mathbf{Z}/p\mathbf{Z}$ if and only if $e > 1$. (Recall that the vanishing of a discriminant is independent of the choice of basis.)

Suppose $e > 1$. Then any $x \in \mathfrak{p} - \mathfrak{p}^e$ is a nonzero nilpotent element in $\mathcal{O}_K/\mathfrak{p}^e$. By linear algebra over fields, such an $\overline{x}$ can be used as part of a $\mathbf{Z}/p\mathbf{Z}$-basis of $\mathcal{O}_K/\mathfrak{p}^e$, say $\{\overline{x}_1, \ldots, \overline{x}_n\}$ with $\overline{x} = \overline{x}_1$. Writing the trace map $\mathrm{Tr}_{(\mathcal{O}_K/\mathfrak{p}^e)/(\mathbf{Z}/p\mathbf{Z})}$ as Tr for short, the first column of the matrix $(\mathrm{Tr}(\overline{x}_i\overline{x}_j))$ contains the numbers $\mathrm{Tr}(\overline{x}_i\overline{x})$. These traces are all $\overline{0}$: $\overline{x}_i\overline{x}$ is nilpotent, so the linear transformation $m_{\overline{x}_i\overline{x}}$ on $\mathcal{O}_K/\mathfrak{p}^e$ is nilpotent and thus its eigenvalues all equal zero. Since one column of the trace-pairing matrix $(\mathrm{Tr}(\overline{x}_i\overline{x}_j))$ is all $\overline{0}$, $\mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/\mathfrak{p}^e) = \overline{0}$.

Now suppose $e = 1$. Then $\mathcal{O}_K/\mathfrak{p}^e = \mathcal{O}_K/\mathfrak{p}$ is a finite field of characteristic p. We want to prove $\mathrm{disc}_{\mathbf{Z}/p\mathbf{Z}}(\mathcal{O}_K/\mathfrak{p}) \neq \overline{0}$. If this discriminant is $\overline{0}$, then (because $\mathcal{O}_K/\mathfrak{p}$ is a field) the trace function $\mathrm{Tr}\colon \mathcal{O}_K/\mathfrak{p} \to \mathbf{Z}/p\mathbf{Z}$ is identically zero. However, from the theory of finite fields, this trace function can be written as a polynomial function:

$$\mathrm{Tr}(t) = t + t^p + t^{p^2} + \cdots + t^{p^{r-1}},$$

where $p^r = |\mathcal{O}_K/\mathfrak{p}|$. Since $\mathrm{Tr}(t)$ as a polynomial in t has smaller degree than the size of $\mathcal{O}_K/\mathfrak{p}$, the function $\mathrm{Tr}(t)$ is not identically zero on $\mathcal{O}_K/\mathfrak{p}$. Therefore the discriminant of a finite extension of $\mathbf{Z}/p\mathbf{Z}$ does not equal zero. □

4. The relative case

Ramified primes and discriminants can be defined for an extension of number fields E/F where the base field F need not be $\mathbf{Q}$. This is called the relative case, while the study of number field extensions with base field $\mathbf{Q}$ is called the absolute case.

To define the discriminant of E/F, a direct copying of the definition over $\mathbf{Q}$ is possible if $\mathcal{O}_F$ is PID. In this case $\mathcal{O}_E$ is a free $\mathcal{O}_F$-module. We say a prime π in $\mathcal{O}_F$ is *ramified* in E when the ideal $\pi\mathcal{O}_E$ has a repeated prime ideal factor and is *unramified* otherwise. When $\{\alpha_1, \ldots, \alpha_n\}$ is an $\mathcal{O}_F$-basis of $\mathcal{O}_E$ the number $\mathrm{disc}_{E/F}(\alpha_1, \ldots, \alpha_n)$ is nonzero in $\mathcal{O}_F$ and changing the $\mathcal{O}_F$-basis of $\mathcal{O}_E$ changes this number by a unit square in $\mathcal{O}_F$.

Example 4.1. Let $F = \mathbf{Q}(i)$ and $E = F(\sqrt{-5})$. Then $\mathcal{O}_F = \mathbf{Z}[i]$ and it can be shown that $\mathcal{O}_E = \mathcal{O}_F[\sqrt{-5}] = \mathbf{Z}[i][\sqrt{-5}]$, so $\{1, \sqrt{-5}\}$ is a $\mathbf{Z}[i]$-basis of $\mathcal{O}_E$. Since

$$\mathrm{disc}_{E/F}(1, \sqrt{-5}) = \det \begin{pmatrix} \mathrm{Tr}_{E/F}(1 \cdot 1) & \mathrm{Tr}_{E/F}(1 \cdot \sqrt{-5}) \\ \mathrm{Tr}_{E/F}(\sqrt{-5} \cdot 1) & \mathrm{Tr}_{E/F}(\sqrt{-5}\sqrt{-5}) \end{pmatrix} = \det \begin{pmatrix} 2 & 0 \\ 0 & -10 \end{pmatrix} = -20,$$

the discriminant of E/F is -20 up to a unit square in $\mathbf{Z}[i]$. Another $\mathbf{Z}[i]$-basis of $\mathcal{O}_E$ is $\{i, \sqrt{-5}\}$ and $\mathrm{disc}_{E/F}(i, \sqrt{-5}) = \det\left(\begin{smallmatrix} -2 & 0 \\ 0 & -10 \end{smallmatrix}\right) = 20$, which differs from -20 by a unit square factor $-1 = i^2$.

Numbers in a PID that differ by a unit factor have the same prime factors, so whether or not a prime in $\mathcal{O}_F$ divides $\mathrm{disc}_{E/F}(\alpha_1, \ldots, \alpha_n)$ does not depend on the choice of $\mathcal{O}_F$-basis.

Here is how Theorem 1.3 generalizes to E/F when $\mathcal{O}_F$ is a PID.

Theorem 4.2. *For an extension of number fields E/F where $\mathcal{O}_F$ is a* PID*, a prime π in F ramifies in E if and only if π divides $\mathrm{disc}_{E/F}(\alpha_1, \ldots, \alpha_n)$ where $\{\alpha_1, \ldots, \alpha_n\}$ is an $\mathcal{O}_F$-basis of $\mathcal{O}_E$.*

Proof. Use the method of proof of Section 3 with $\mathbf{Z}$ replaced by $\mathcal{O}_F$. □

Example 4.3. In Example 4.1, 20 has prime factorization $-(1+i)^4(1+2i)(1-2i)$, in $\mathbf{Z}[i]$ so the primes in $\mathbf{Z}[i]$ that ramify in $\mathbf{Z}[i, \sqrt{-5}]$ are $1+i$, $1+2i$, and $1-2i$.

What do we do when $\mathcal{O}_F$ is not necessarily a PID? A (nonzero) prime ideal $\mathfrak{p}$ in $\mathcal{O}_F$, is called *ramified* in E if the ideal $\mathfrak{p}\mathcal{O}_E$ has a repeated prime ideal factor in $\mathcal{O}_E$ and *unramified*

otherwise. It can be shown that when $E = F(\alpha)$ with $\alpha \in \mathcal{O}_E$ and α has minimal polynomial $f(T)$ in $F[T]$, the prime $\mathfrak{p}$ is unramified in E if (but not necessarily only if) $\mathrm{disc}(f(T)) \not\equiv 0 \bmod \mathfrak{p}$. A nonzero number in $\mathcal{O}_F$ has only finitely many prime ideal factors, so all but finitely many prime ideals in $\mathcal{O}_F$ are unramified in E.

The discriminant of E/F is an ideal in $\mathcal{O}_F$, called the *discriminant ideal* $\mathfrak{d}_{E/F}$. It is defined as the ideal in $\mathcal{O}_F$ generated by the numbers $\mathrm{disc}_{E/F}(\alpha_1, \ldots, \alpha_n)$ where $\{\alpha_1, \ldots, \alpha_n\}$ runs over all F-bases of E that are contained in $\mathcal{O}_E$.[3] If one F-basis of E contained in $\mathcal{O}_E$ is in the $\mathcal{O}_F$-span of another F-basis of E contained in $\mathcal{O}_E$, then the discriminant of the first basis is a multiple of the discriminant of the second basis, so if $\mathcal{O}_E$ has an $\mathcal{O}_F$-basis, then $\mathfrak{d}_{E/F}$ is a principal ideal with the discriminant of any $\mathcal{O}_F$-basis of $\mathcal{O}_E$ being a generator.

Theorem 1.3 has the following generalization to the relative case, whose proof is more technical than that of Theorem 1.3 and is omitted.

Theorem 4.4. *For an extension of number fields E/F, a prime $\mathfrak{p}$ in F ramifies in E if and only if $\mathfrak{p}$ divides the discriminant ideal $\mathfrak{d}_{E/F}$.*

References

[1] R. Dedekind, *Über den Zusammenhang zwischen der Theorie der ideale und der Theorie der höheren Congruenzen*, Abhandlungen der Königlichen Gesellschaft der Wissenschaften in Göttingen **23** (1878), 3–37. URL https://eudml.org/doc/135827.

[2] R. Dedekind, *Über die Discriminanten endlicher Körper*, Abhandlungen der Königlichen Gesellschaft der Wissenschaften in Göttingen **29** (1882), 1–56. URL https://eudml.org/doc/135869.

[3] The ideal $\mathfrak{d}_{E/F}$ can be defined in a second way using localizations: see https://math.stackexchange.com/questions/4875752/. Some work is needed to show this way of defining $\mathfrak{d}_{E/F}$ agrees with the first way presented above.